ITHERE TECHNOLOGY GROUP

ITHERE CARE

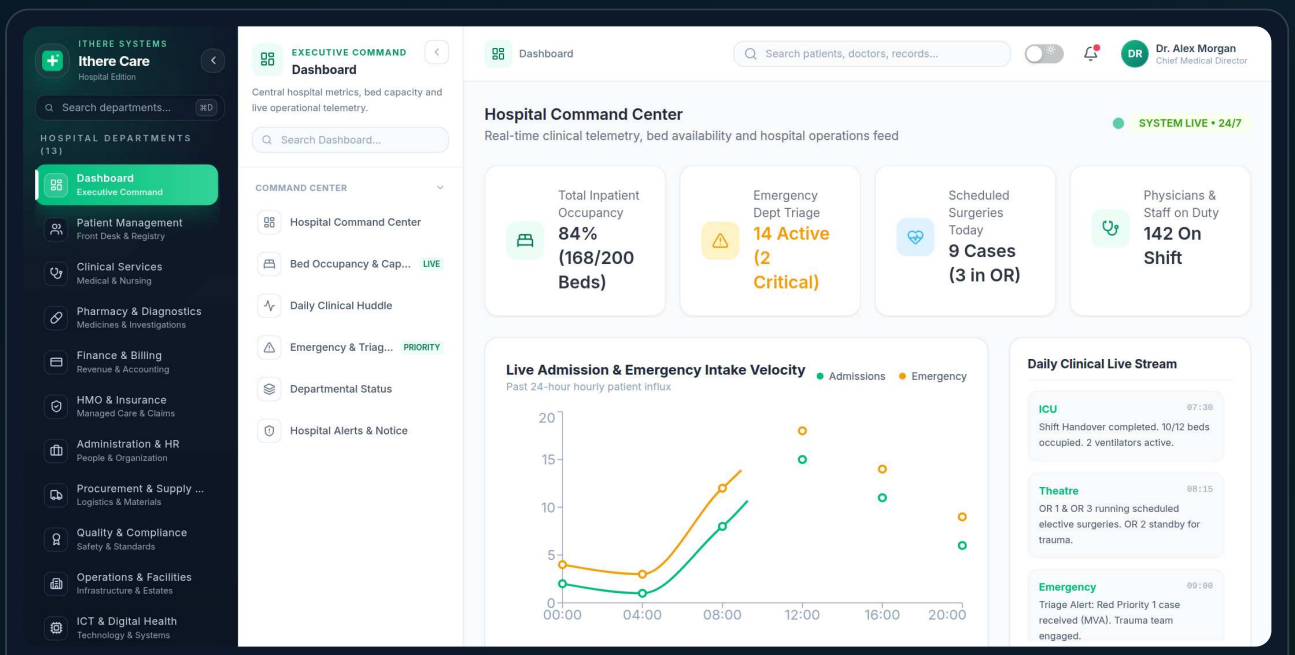
OFFICIAL SYSTEM DOCUMENTATION

INTEGRATED ENTERPRISE HEALTHCARE SOLUTION

ITHERE SYSTEMS
HOSPITAL EDITION

Powered by Ithere Care — Healthcare Management Information System

Complete Application Walkthrough, Brand Architecture Standard, Departmental Functional Inventory (13 Divisions / 169 Views), Technical Architecture Blueprint, API & Data Model Specification, and Ecosystem Readiness Reference.



Live Workspace Capture — Ithere Systems (Hospital Edition) • Executive Command Center & Two-Tier Departmental Navigation

PRIMARY PRODUCT **Ithere Care (HMIS / EMR)**	DEPLOYMENT IDENTITY **Ithere Systems — Hospital Edition**	APPLICATION VERSION **v0.1.0 (Development)**
PARENT ORGANIZATION **Ithere Technology Group**	ARCHITECTURE SCOPE **13 Departments • 169 Views**	DATE PREPARED **November 2026 (Baseline ae06a5b)**

00 Document Control & Table of Contents

Document Title	**Ithere Systems — Hospital Edition: Comprehensive System Documentation & Functional Inventory**
Primary Product Identity	**Ithere Care** (Healthcare Management Information System & EMR)
Deployment Solution Name	**Ithere Systems — Hospital Edition** (*Powered by Ithere Care*)
Parent Company	**Ithere Technology Group** — *"The Future Has an Address."*
Official Application Version	`0.1.0` (Displayed as `v0.1.0 (Development)` / `v0.1.0-dev`)
Codebase Baseline Inspected	Branch `refactor` (Commit `ae06a5b`) • React 19.2 + TypeScript 5.9 + Vite 7.3

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01 Document Overview

This document serves as the official technical, functional, and architectural reference for **Ithere Care**, deployed as an integrated enterprise healthcare solution under **Ithere Systems — Hospital Edition**. It is based on a direct, line-by-line inspection of the existing source code, routing table, state stores, API services, and UI screens.

> **STRICT VERIFICATION & HONESTY STANDARD**
>
> Every module across all 13 hospital departments (169 total views) has been verified against its actual TypeScript implementation. Modules are explicitly categorized into: **(1) Fully Implemented (API Wired)**, **(2) Partially Implemented**, **(3) UI Only (Interactive Local State)**, or **(4) Planned / UI Staging Blueprint (HmisPlaceholderPage)**. No unimplemented functionality or unverified compliance certification is claimed as complete.

02 Brand and Product Overview

Ithere Technology Group (*"The Future Has an Address."*) operates under a unified multi-product strategy: **One platform. Multiple independent products. Shared infrastructure. Flexible subscriptions.**

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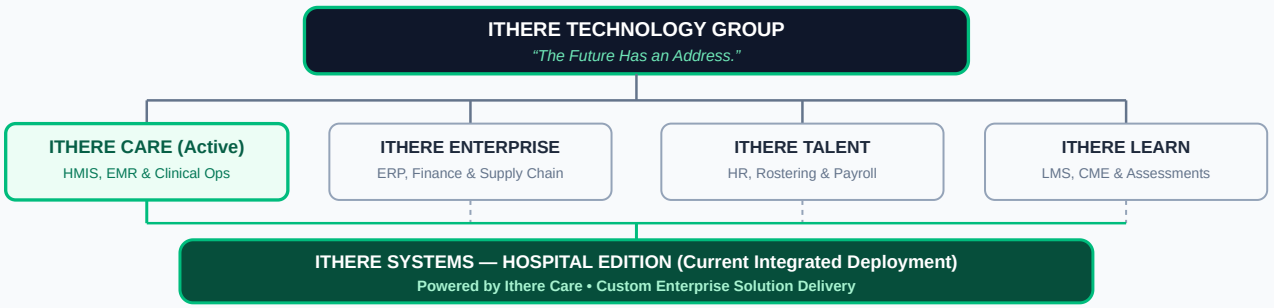


Figure 2.1 — Ithere Technology Group Brand Hierarchy & Product Ecosystem Positioning

BRAND ENTITY	ROLE IN HIERARCHY	SCOPE & REPRESENTATION IN CURRENT CODEBASE
Ithere Technology Group	Parent Company	Corporate owner displayed in Login hero badge, MainSidebar copyright footer, and System Administration profile. Tagline: *"The Future Has an Address."*
Ithere Care	Primary Healthcare Product	The core HMIS/EMR application in this repository. Owns all clinical, patient, laboratory, diagnostic, pharmacy, and HMO workflows.
Ithere Systems — Hospital Edition	Integrated Deployment Identity	Communicates that this hospital installation packages **Ithere Care** together with enterprise hospital operations (Finance, HR, Procurement, ICT).
Ithere Enterprise / Talent / Learn	Future Standalone Products	Not yet separate external products; their hospital-specific counterparts (Finance/Procurement, HR/Payroll, CME Training) exist as modular department routes inside the Hospital Edition.

03 Version Information & Standardization

An inspection of version metadata across the repository prior to rebranding found: (1) `package.json` had `"version": "0.0.0"` (Vite template default), (2) Git history contained 16 commits on `main` / `refactor` with **zero Git tags**, and (3) `SystemConfiguration.tsx` had a static mock string `"v2.4.0 Release"`.

To establish a single, truthful engineering baseline, version **`0.1.0`** (Status: **`Development`**, formatted as **`v0.1.0 (Development)`**) has been established in `src/app/appInfo.ts` and `package.json`.

FILE / COMPONENT	CONSTANT / FIELD USED	RENDERED UI / METADATA OUTPUT
`src/app/appInfo.ts`	`APP_INFO.version` / `versionLabel`	Single source of truth: `version: "0.1.0"`, `versionLabel: "v0.1.0 (Development)"`, `shortVersion: "v0.1.0-dev"`
`package.json`	`"version"`	Updated from `"0.0.0"` to `"0.1.0"`
`src/pages/auth/Login.tsx`	`APP_INFO.versionLabel`	Login hero footer & form subtitle display `Powered by Ithere Care • v0.1.0 (Development)`
`src/components/navigation/MainSidebar.tsx`	`APP_INFO.shortVersion`	Sidebar footer displays `Ithere Systems — Hospital Edition` / `Powered by Ithere Care • v0.1.0-dev`

FILE / COMPONENT	CONSTANT / FIELD USED	RENDERED UI / METADATA OUTPUT
`src/components/Header.tsx`	`APP_INFO.shortVersion`	User profile flyout displays `Ithere Care • v0.1.0-dev`
`SystemSettings.tsx` & `SystemConfiguration.tsx`	`APP_INFO.versionLabel`	System Administration & ICT KPI cards display `v0.1.0 (Development)`

04 Executive Summary of the Application

HOSPITAL DIVISIONS
13 Depts
Two-Tier Navigation Rail

TOTAL ROUTE VIEWS
169 Views
Registered in AppRoutes.tsx

LIVE API FLOWS
9 Views
Auth, Patients & HMOs

RICH INTERACTIVE UI
16 Views
Command, Lab & Pharmacy

Ithere Care is an enterprise Hospital Management Information System (HMIS) frontend application built with React 19, TypeScript, Tailwind CSS, and Ant Design 6. Deployed as **Ithere Systems — Hospital Edition**, it provides a unified clinical and administrative workspace for multispecialty hospitals, covering patient registration, outpatient/inpatient care, laboratory diagnostics, pharmacy operations, revenue cycle, HMO insurance management, human resources, supply chain, quality compliance, and ICT governance.

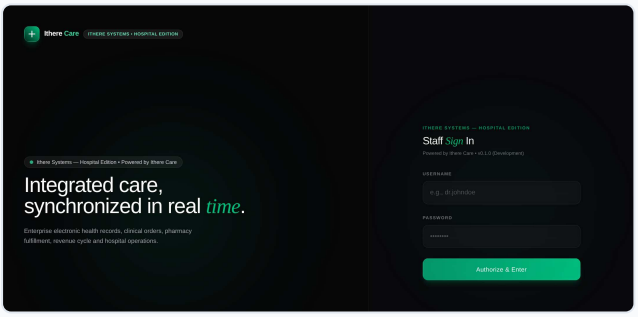


Figure 4.1 — Rebranded Login Screen (Ithere Care • Ithere Systems Hospital Edition)

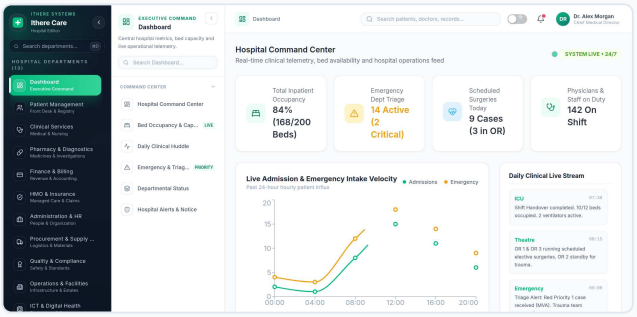


Figure 4.2 — Hospital Enterprise Command Center & Two-Tier Departmental Sidebar

05 System Architecture Overview

The application is architected as a modular single-page application (SPA) with a strict separation between routing (`src/routes/`), navigation metadata (`src/navigation/`), presentation layouts (`src/layouts/`), state stores (`src/store/`), and REST API adapters (`src/services/`).

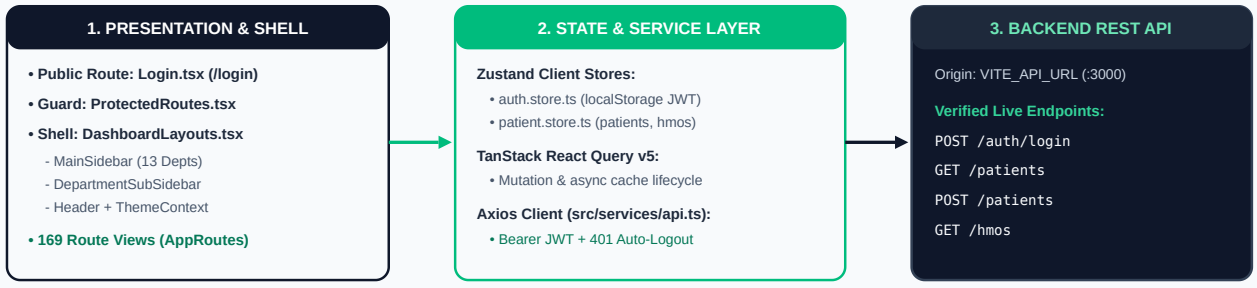


Figure 5.1 — Ithere Care Three-Tier Frontend-to-Backend Runtime Architecture

06 User Roles and Access Control

Authentication is managed through a combination of TanStack React Query mutations (`src/features/auth/authApi.ts`), Zustand persisted state (`src/store/auth.store.ts`), React Router route guards (`src/routes/ProtectedRoutes.tsx`), and Axios HTTP interceptors (`src/services/api.ts`).

- **Session Persistence:** Upon successful `POST /auth/login`, `setAuth(user, token)` persists the staff profile (`id`, `name`, `email`, `role`) and JWT `token` to `localStorage` under `"auth-storage"`.
- **Route Guarding:** `ProtectedRoutes.tsx` wraps the entire `DashboardLayout` route tree. Unauthenticated visitors are redirected to `/login`.
- **Automatic 401 Session Revocation:** If the backend returns HTTP `401 Unauthorized`, the Axios response interceptor in `src/services/api.ts` automatically executes `useAuthStore.getState().logout()` and redirects to `/login`.
- **Current RBAC Limitation:** While user roles are displayed in the UI header and sidebar (`user?.role || "Administrator"`) and role management screens exist at `/system-admin/permissions` (`RolePermissions.tsx`), granular per-route RBAC enforcement is not yet active on the frontend router.

07 Complete Application Walkthrough

The application is organized around a two-tier departmental navigation system defined in `src/navigation/navigationConfig.ts`. Below is a walkthrough of the live workspaces captured directly from the rebranded build.

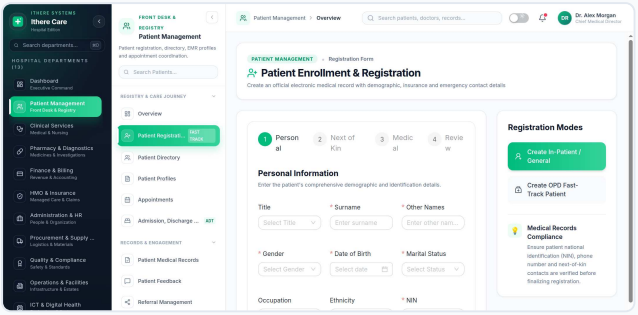


Figure 7.1 — Patient Registration Hub (/patients/registration) — Inpatient & OPD Intake Flows

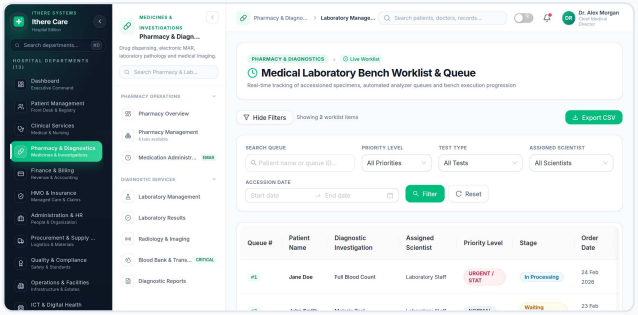


Figure 7.2 — Medical Laboratory Active Queue (/lab/queue) — Sample & Processing Status

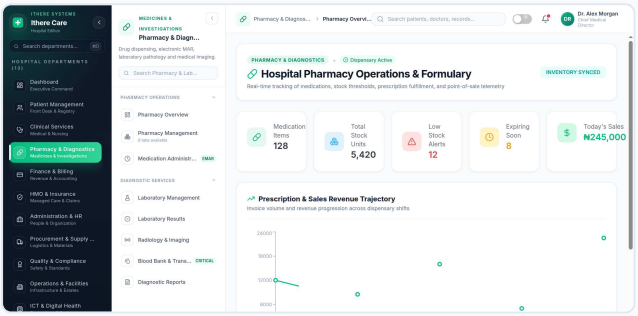


Figure 7.3 — Pharmacy & Dispensary Dashboard (/pharmacy/overview) — Stock & Dispensing

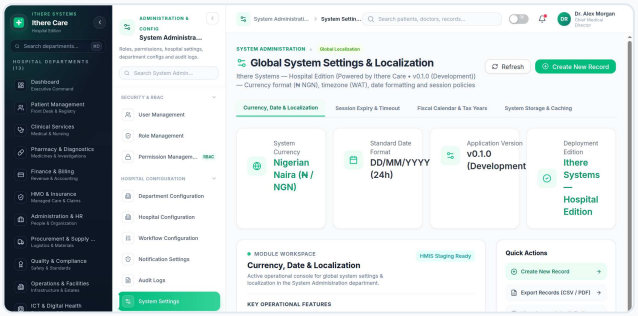


Figure 7.4 — Global System Settings (/system-admin/settings) — Edition & Version v0.1.0

Walkthrough of Core Interactive & Staged Workspaces

- **Command Center (`/`):** `HospitalDashboard.tsx` renders 6 hospital-wide KPI cards, a 12-department quick-launch matrix, critical clinical/operational alerts (ICU, Blood Bank, HMO Claims, Cold Chain), and ward occupancy progress bars.
- **Patient Management (`/patients/*`):** `Patients.tsx` (`/patients/list`) connects to `GET /patients` with debounced search, category/status filters, and server pagination, opening `PatientProfile.tsx` and `OpdPatientProfile.tsx` modals. `CreatePatientDashboard.tsx` (`/patients/create`) hosts both the multi-step category registration (`CreateNewPatient.tsx` , wired to `POST /patients` and `GET /hmos`) and outpatient intake (`CreateOPDPatient.tsx`).
- **Medical Laboratory (`/lab/*`):** 6 interactive screens (`LabDashboard` , `NewLabRequests` , `LabRequests` , `LabQueue` , `LabResults` , `LabReports`) plus `LabResultModal.tsx` (parameter result entry) and `LabReportModal.tsx` (printable clinical report with reference ranges).
- **Pharmacy & Dispensary (`/pharmacy/*`):** 8 interactive screens (`PharmacyDashboard` , `Medications` , `Inventory` , `Prescriptions` , `PharmacyOrders` , `Sales` , `Suppliers` , `PharmacyManagement`) supporting formulary search, batch stock tracking, prescription dispensing, and POS sales in local state, plus `EmarDispensing.tsx` .
- **Enterprise Department Blueprints (`HmisPlaceholderPage.tsx`):** Used across 144 sub-modules in Clinical Services, Diagnostics, Finance, HMO, HR, Procurement, Quality, Operations, ICT, Reports, and System Admin to provide consistent departmental KPIs, interactive sub-tab switching, search bars, and capability checklists.

08 Module-by-Module Functional Breakdown

#	DEPARTMENT	ROUTE PREFIX	VIEWS	IMPLEMENTATION STATUS	VERIFIED FUNCTIONAL SCOPE & KEY COMPONENTS
1	**Command Center**	`/`, `/dashboard/*`	6	1 UI MOCK 5 STAGED	`HospitalDashboard.tsx` (interactive command center) + 5 executive/clinical/occupancy/revenue sub-views.
2	**Patient Management**	`/patients/*`	11	4 LIVE API 1 UI MOCK 6 STAGED	`Patients.tsx` (`GET /patients`), `CreatePatientDashboard.tsx`, `CreateNewPatient.tsx` (`POST /patients`, `GET /hmos`), `CreateOPDPatient.tsx`, `PatientDashboard.tsx`, plus Admissions, Referrals, Discharges, Records, Vital Registry, Consents.
3	**Clinical Services**	`/appointments`, `/clinical/*`	17	1 PARTIAL API 16 STAGED	`appointments/Home.tsx` (reads `usePatientStore`) + 16 clinical specialty modules (OPD, Triage, Consultation, IPD, EMR, Nursing, ICU, Emergency, Theatre, Maternity, ANC, Physio, Dietary, Dental, Ophthalmology, Mental Health).
4	**Diagnostics & Allied**	`/lab/*`, `/diagnostics/*`	10	6 UI MOCK 4 STAGED	6 rich Lab screens (`LabDashboard`, `NewLabRequests`, `LabRequests`, `LabQueue`, `LabResults`, `LabReports` + 2 modals) + 4 staged Diagnostic views (Radiology PACS, CT/MRI, Blood Bank, Pathology).
5	**Pharmacy & Dispensary**	`/pharmacy/*`	9	8 UI MOCK 1 STAGED	8 interactive Pharmacy screens (`PharmacyDashboard`, `Medications`, `Inventory`, `Prescriptions`, `PharmacyOrders`, `Sales`, `Suppliers`, `PharmacyManagement`) + staged `EmarDispensing`.
6	**Finance & Revenue**	`/finance/*`	11	11 STAGED	Finance Overview, Patient Billing, Cashier POS, Inpatient Deposits, Service Tariff Book, General Ledger, AR, AP, Doctor Honorarium, Budgeting, Financial Statements.
7	**HMO & Insurance**	`/hmo/*`	12	12 STAGED	HMO Command, Providers Directory, NHIA Schemes, Corporate Retainers, Eligibility Check, Pre-Auth, Tariffs & Capitation, Claims Scrubbing, Batch Submission, Remittance, Denials, Utilization. (Note: `GET /hmos` service is live for Patient Registration).

#	DEPARTMENT	ROUTE PREFIX	VIEWS	IMPLEMENTATION STATUS	VERIFIED FUNCTIONAL SCOPE & KEY COMPONENTS
8	**Administration & HR**	`/hr/*`	16	16 STAGED	HR Overview, Staff Directory, Medical Credentialing, Shift Rostering, Attendance, Leave, Payroll, Allowances, Tax/Pension, Recruitment, Appraisals, CME Training, Disciplinary, Memos, Visitors, Mortuary Admin.
9	**Procurement & Supply**	`/procurement/*`	15	15 STAGED	Supply Chain Overview, Requisitions, RFQ/Tenders, Purchase Orders, GRN, Central Store, Sub-Stores, Stock Transfers, Surgical Kits, Vendors, SLA Contracts, Asset Registry, Preventive Maint., Batch Tracking, Stock Audit.
10	**Quality & Compliance**	`/quality/*`	13	13 STAGED	Quality Overview, Incident Reporting, Morbidity & Mortality, Infection Control, Clinical Audit, Pharmacovigilance, Patient Feedback, Accreditation, SOP Library, Risk Register, Ethics Audit, OHS, Regulatory Returns.
11	**Operations & Facilities**	`/operations/*`	14	14 STAGED	Operations Command, Housekeeping, Laundry/Linen, CSSD Sterilization, Biomedical Eng., Ambulance Dispatch, Medical Gas/Oxygen, Power/HVAC, Medical Waste, Catering, Security/CCTV, Access Control, Work Orders, Emergency Drills.
12	**ICT & Digital Health**	`/ict/*`	12	12 STAGED	ICT Overview, System Configuration (`v0.1.0`), HL7/FHIR/PACS, LIS/RIS Middleware, Telemedicine, Mobile App/SMS, Biometrics, Network/Server Health, Backup/DR, Cybersecurity, IT Service Desk, API Keys.
13	**Reports & System Admin**	`/reports/*`, `/system-admin/*`	19	19 STAGED	10 Enterprise Analytics & NHMIS Reporting views + 9 System Admin views (Users, RBAC Permissions, Departments, Ward/Bed Master, Alerts, Audit Logs, `HospitalConfig.tsx`, `SystemSettings.tsx`).

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09 Core Hospital Workflows



Figure 9.1 — End-to-End Hospital Patient Journey Across Ithere Care Modules

10 Database & Data Model Summary

The frontend TypeScript interfaces in `src/types/patient.d.ts`, `src/store/auth.store.ts`, and `src/services/hmo.service.ts` define the canonical domain schemas exchanged with the backend database:

ENTITY	SOURCE FILE	KEY ATTRIBUTES & RELATIONSHIPS
User	`src/store/auth.store.ts`	`id: string`, `name: string`, `email: string`, `role?: string`. Authenticated via JWT Bearer token.
Patient	`src/types/patient.d.ts`	`id: number`, `cardNumber: string`, `firstName`, `surname`, `otherName`, `gender`, `dob`, `phoneNumber`, `email`, `address`, `bloodGroup`, `genotype`, `category` (`General | Private | Corporate | Insurance | Family | OPD`), `status` (`Active | Inactive | Pending`), Next-of-Kin fields, `hmoId` (FK), `hmoNumber`, `companyName`, `employeeId`, `familyHeadId`.
Hmo	`src/services/hmo.service.ts`	`id: string | number`, `name: string`, `code?: string`, `status?: string`. Referenced by `Patient.hmoId` for Insurance enrollees.

11 API and Backend Overview

All HTTP communication is routed through the centralized Axios instance in `src/services/api.ts` using `baseURL: import.meta.env.VITE_API_URL` (configured as `http://localhost:3000` in `.env`).

METHOD	ENDPOINT	FRONTEND SERVICE	REQUEST / RESPONSE CONTRACT
`POST`	`/auth/login`	`loginUser` (`authApi.ts`)	**Req:** `{ username, password }` → **Res:** `{ data: { user, access_token } }`
`GET`	`/patients`	`PatientService.getPatients`	**Query:** `?page&limit&search&category&status` → **Res:** Paginated `Patient[]` + `meta`
`POST`	`/patients`	`PatientService.createPatient`	**Req:** Multi-category `Partial<Patient>` payload → **Res:** Normalized `Patient` record
`GET`	`/hmos`	`HmoService.getHmos`	**Req:** None → **Res:** Accredited `Hmo[]` list for insurance registration dropdown

12 Frontend Structure & 13 Configuration Overview

- **Brand & Version Constants (`src/app/appInfo.ts`):** Exports `APP_INFO` as the single source of truth for `Ithere Care`, `Ithere Systems — Hospital Edition`, `Ithere Technology Group`, and `v0.1.0 (Development)`.
- **Navigation Registry (`src/navigation/navigationConfig.ts`):** Declaratively defines all 13 departments (`HMIS_DEPARTMENTS`), icons, categories, and sub-routes.
- **Theme & Design System (`src/theme/themeConfig.ts` & `tailwind.config.js`):** Defines the clinical emerald primary palette (`#00bc7d`), dark-mode surfaces, and Ant Design 6 tokens.
- **Environment Configuration (`.env`):** Configures `VITE_API_URL=http://localhost:3000`.

14 Current Implementation Status & 15 Known Limitations

> **TRANSPARENT ENGINEERING ASSESSMENT (VERSION 0.1.0 — DEVELOPMENT)**
>
> Out of **169 total route views** in the application:
>
> - **9 views/flows are connected to live backend REST endpoints** (Authentication, Patient Directory, Multi-Category Inpatient Registration, OPD Registration, and HMO Lookup).
> - **16 views (+ 4 modals) are rich, interactive frontend implementations** operating on local state (Hospital Command Center, Patient Overview, 6 Medical Laboratory screens, and 8 Pharmacy screens).
> - **144 views are structured UI staging shells** rendered via `HmisPlaceholderPage.tsx`, providing department-specific KPIs, tabs, and workflow specifications awaiting backend API endpoints.

Key Pending Engineering Items: (1) Backend REST services for Clinical Consultations/EMR, Lab orders/results persistence, Pharmacy inventory/POS persistence, Finance/Billing, HMO Claims, HR, and Procurement; (2) Fine-grained route-level RBAC enforcement in `ProtectedRoutes.tsx`; (3) Multi-organization tenant context headers (`X-Organization-Id`) for multi-hospital cloud deployments.

16 Future Ecosystem Readiness

The modular department structure in `src/navigation/navigationConfig.ts` and `src/pages/*` aligns directly with the long-term **Ithere Technology Group** multi-product roadmap:

- **Ithere Care:** Retains all clinical, patient, diagnostic, laboratory, pharmacy, and HMO modules.
- **Ithere Enterprise Integration:** The isolated `/finance/*` (11 views) and `/procurement/*` (15 views) namespaces can progressively bind to shared **Ithere Enterprise** ERP & accounting microservices.
- **Ithere Talent Integration:** The isolated `/hr/*` (16 views) namespace provides a clean boundary for **Ithere Talent** workforce, rostering, and payroll services.
- **Ithere Learn Integration:** The `/hr/training` CME & Training module connects naturally to **Ithere Learn**.
- **Ithere Systems Subscription Control:** Because the 13 departments are rendered dynamically from `HMIS_DEPARTMENTS`, customer-specific module entitlements can filter the navigation rail per hospital subscription tier.

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17 Appendix — Rebranding & Version Changes Applied

FILE PATH	PREVIOUS VALUE / BRANDING	NEW OFFICIAL ITHERE BRANDING & VERSION
`src/app/appInfo.ts` *(Created)*	*Did not exist*	Canonical `APP_INFO` (`Ithere Care` , `Ithere Systems — Hospital Edition` , `v0.1.0`)
`public/ithere-icon.svg` *(Created)*	`/vite.svg`	Custom emerald medical cross vector icon (`#00bc7d`)
`package.json`	`"version": "0.0.0"`	`"version": "0.1.0"`
`index.html`	`<title>MyClinical</title>` , `/vite.svg`	`<title>Ithere Care — Ithere Systems Hospital Edition</title>` , `/ithere-icon.svg`
`src/pages/auth/Login.tsx`	`myClinical` , `Enterprise HMIS` , unverified compliance claim	`Ithere Care` , `Ithere Systems • Hospital Edition` , `Powered by Ithere Care • v0.1.0 (Development)` , `Ithere Technology Group — The Future Has an Address.`
`src/components/navigation/MainSidebar.tsx`	`Enterprise HMIS` / `myClinical` , `© 2026 myClinical HMIS`	`Ithere Systems` / `Ithere Care` (`Hospital Edition`), `Powered by Ithere Care • v0.1.0-dev` , `© 2026 Ithere Technology Group`
`src/components/Header.tsx`	`alex.clinic@myclinical.health`	`alex.clinic@itherecare.health` + `Ithere Care • v0.1.0-dev` badge
`src/components/Sidebar.tsx`	`Clinical Suite` / `MyClinic`	`Ithere Systems` / `Ithere Care` , `© 2026 Ithere Care • v0.1.0-dev`
`src/pages/system-admin/HospitalConfig.tsx`	`myClinical Multispecialty`	`Ithere Systems — Hospital Edition` , `Powered by Ithere Care (v0.1.0-dev)` , `Ithere Technology Group`
`src/pages/system-admin/SystemSettings.tsx`	Generic localization metrics	Displays `Application Version: v0.1.0 (Development)` & `Deployment Edition: Ithere Systems — Hospital Edition`
`src/pages/ict/SystemConfiguration.tsx`	`v2.4.0 Release` (mock string)	`v0.1.0 (Development)` via `APP_INFO.versionLabel`